

Abstract Machine Hierarchy - L0 & L1

Level 5 - 0 (see part notes)

Level 0 - Digital logic
- 1s and 0
- 5V and 0V

◦ Data Bus

- Carries data

◦ Instruction Bus

- carries instructions

◦ Control Bus

- Carries control signals

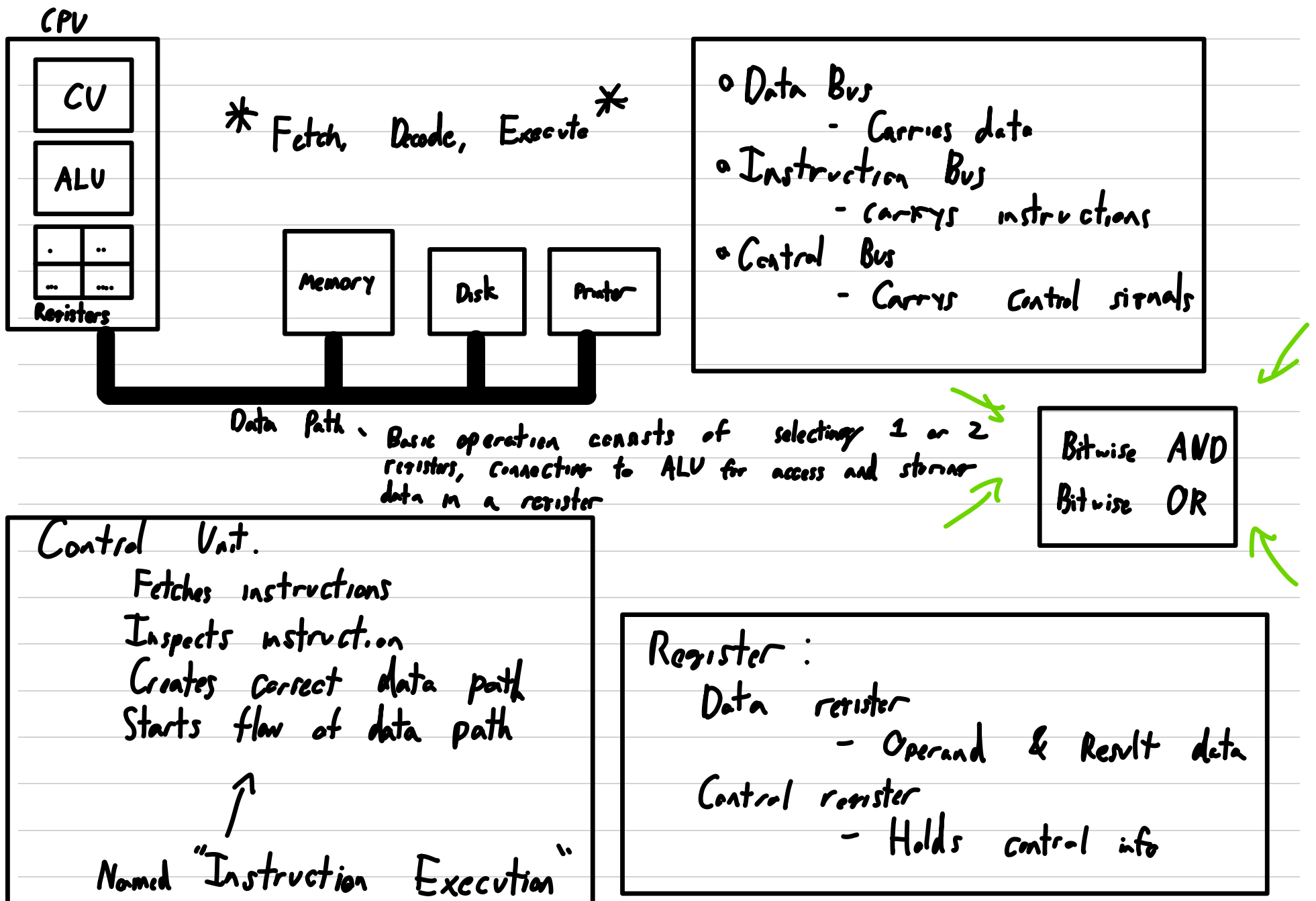
L1 - Microarchitecture

Level 1 - Microarchitecture

- Components comprised of L0
- Registers, CU, ALU, Data path
- 32 bit register is 32 bits wide
- 64 bit registers are 64 bits wide

Register - holds small amount of data

* Carpenter store analogy is good * (8 min into recording)
(1 box of nails vs 1 nail)



Arithmetic + Logic Unit

- Transforms input data to output data with $\overset{\text{arithmetic}}{+, -, *, /}$ or logic $\rightarrow$ {AND, OR, NOT, Shift and Rotate}

L2 - ISA (Instruction set architecture) level

Defines machine code instructions

which either (a) directly controls hardware circuits @ L0
(b) are interpreted @ L1

Instructions are defined by a manufacturer's "MRM"
(Machine Reference Manual)

L3 - Operating System Machine Level

Hybrid level containing L2 and L3 instructions

Contains:

- a particular memory organisation
- ability to run 2 or more programs concurrently

L4 - Assembly

- Symbolic form for one of the underlying languages.
- Provides a method for system programmers to write programs for levels 1, 2 and 3
- Assembly is translated (by an assembler) in order to run on either layer 3, 2 or 1

* Starting at level 4, programming languages contain words or abbreviations *

L5 - Problem Oriented language level

Level consists of languages for application programmers

- High-level languages

- C

- C++

- Java

- Perl

- Python

- PHP